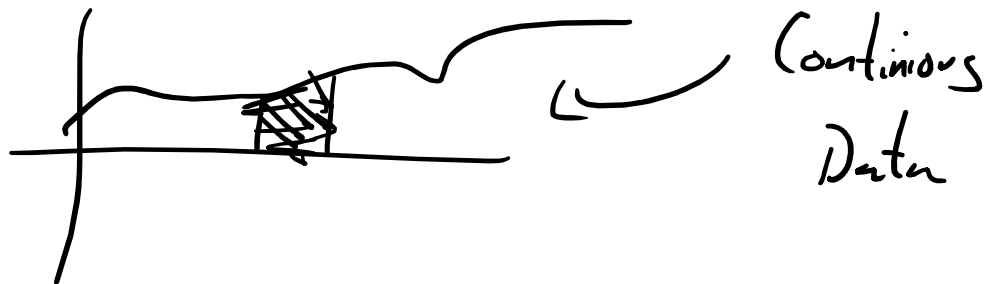


$\Rightarrow$ Mod 4: Summarizing Data:

$\Rightarrow$ Probability Density Function:



$\Rightarrow$ A function $f(x)$ is PDF

if i) $f(x) \geq 0$; $-\infty < x < \infty$

$$\text{ii) } \int_{-\infty}^{\infty} f(x) dx = 1$$

$\Rightarrow$ If x is a CRV with following PDF:

$$f(x) = \begin{cases} \alpha(2x - x^2) & ; 0 < x < 2 \\ 0 & \text{otherwise.} \end{cases}$$

find i) α ii) $P(X > 1)$

$\Rightarrow$ Solve by definition of PDF $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\Rightarrow \int_{-\infty}^0 \cancel{f(x)} dx + \int_0^2 f(x) dx + \int_2^{\infty} \cancel{f(x)} dx = 1$$

$$\Rightarrow \int_0^2 \alpha (2x - x^2) dx = 1$$

$$\Rightarrow \alpha \left[x^2 - \frac{x^3}{3} \right]_0^2 + C = 1$$

$$\Rightarrow \alpha \left[4 - \frac{8}{3} \right] = 1$$

$$\Rightarrow \alpha \frac{4}{3} = 1$$

$$\therefore \boxed{\alpha = \frac{3}{4}}$$

$$P(X > 1) = \int_1^2 \frac{3}{4} (2x - x^2) dx$$

$$= \frac{3}{4} \left[x^2 - \frac{x^3}{3} \right]_1^2$$

$$= \frac{3}{4} \left\{ (4-1) - \left(\frac{8}{3} - \frac{1}{3} \right) \right\}$$

$$= \frac{3}{4} \left(3 - \frac{7}{3} \right) = \frac{3}{4} \times \frac{2}{3} = \boxed{\frac{1}{2}}$$

$$\Rightarrow f(x) = \begin{cases} kx^2 & -3 < x < 3 \\ 0 & \text{otherwise} \end{cases}$$

Find k , $P(1 \leq x \leq 2)$; $P(x \leq 2)$; $P(x > 1)$

$$\Rightarrow \int_{-3}^3 kx^2 dx = 1 \quad \boxed{\because \text{PDF } f(\cdot) = 1}$$

$$\Rightarrow 2k \int_0^3 x^2 dx = 1 \quad ;$$

$$\Rightarrow 2k \left(\frac{x^3}{3} \right)_0^3$$

$$\Rightarrow 2k \left[\frac{27}{3} \right] = 1$$

$$\Rightarrow 18k = 1 \quad \therefore \boxed{k = \frac{1}{18}}$$

$$\Rightarrow P(1 \leq x \leq 2) = \int_1^2 f(x) dx = 1$$

$$\begin{aligned} \Rightarrow \frac{1}{18} \int_1^2 x^2 dx &= \frac{1}{18} \left[\frac{x^3}{3} \right]_1^2 \\ &= \frac{1}{18} \left[\frac{8}{3} - \frac{1}{3} \right] \\ &= \frac{1}{18} \left[\frac{7}{3} \right] \\ &= \frac{7}{54} \end{aligned}$$

$$\begin{aligned} P(x \leq 2) &= \int_{-3}^2 f(x) dx = 1 \\ &= \frac{1}{18} \left(\frac{x^3}{3} \right)_{-3}^2 = \frac{1}{18} \left(\frac{8}{3} - \frac{-27}{3} \right) \\ &= \frac{1}{18} \left[\frac{8}{3} + \frac{27}{3} \right] \\ &= \frac{1}{18} \left[\frac{35}{3} \right] \\ &= \boxed{\frac{35}{54}} \end{aligned}$$

$$\Rightarrow P(x > 1)$$

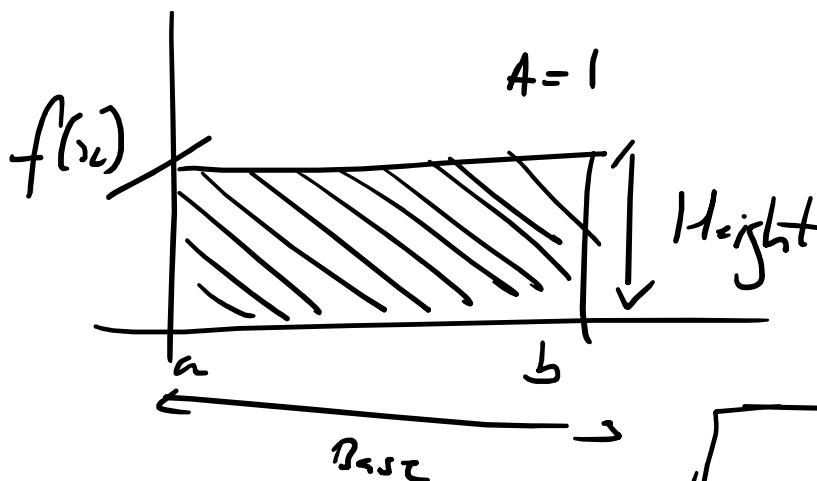
$$= \frac{1}{18} \int_1^3 x^2 dx = 1 \Rightarrow \frac{1}{18} \left[\frac{x^3}{3} \right]_1^3$$

$$= \frac{1}{18} \left[\frac{27}{3} - \frac{1}{3} \right]$$

$$= \frac{1}{18} \left[\frac{26}{3} \right] = \boxed{\frac{26}{54}}$$

$$= \boxed{\frac{13}{27}}$$

$\Rightarrow$ Uniform Distⁿ problems:

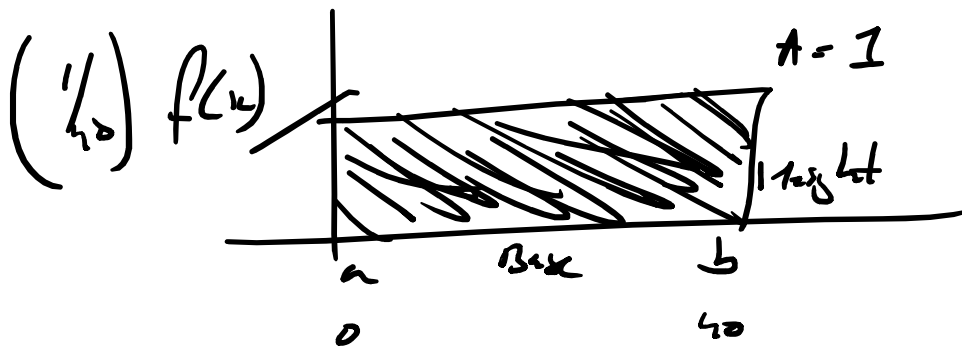


$$A=1 : f(x) \quad \therefore \boxed{f(x) = \frac{1}{b-a}}$$

$$A = 1 : f(x) = \frac{1}{b-a}$$

$$\boxed{A = \text{Base} \times \text{Height}}$$

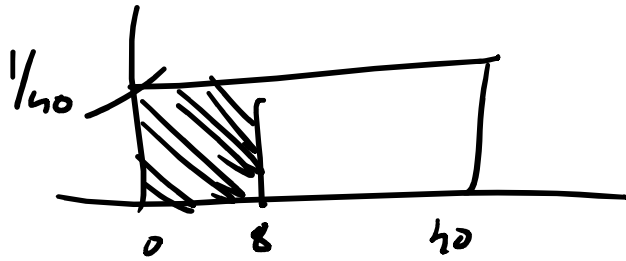
$\Rightarrow$ The amt of time a person must wait for a train to arrive in a certain town is uniformly distributed b/w 0 & 40 mins.



$$f(x) = \frac{1}{b-a} = \frac{1}{40}$$

$$A = B \times H : 40 \times \frac{1}{40} = 1 \quad \textcircled{1}$$

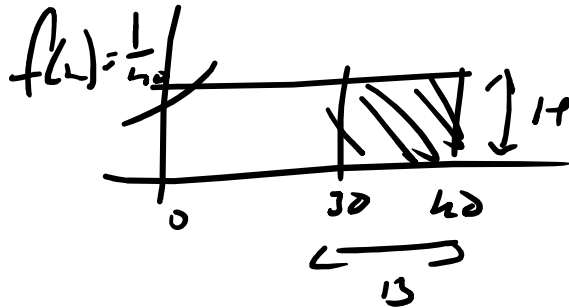
$\Rightarrow$ What is the probability that a person must wait less than 8 mins?



$$P(X < 8) = ?$$

$$A = BH = 8 \times \frac{1}{40} = \frac{1}{5} \approx \underline{\underline{20\%}}$$

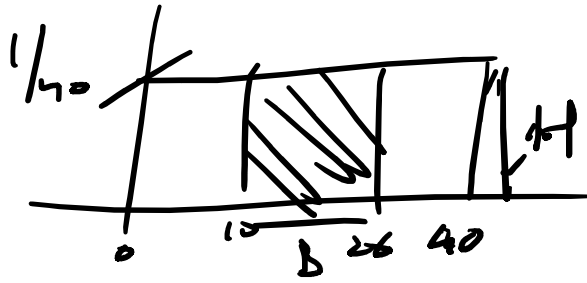
$\Rightarrow$ What is the probability that a person must wait more than 30 mins?



$$P(X > 30) = P(30 < X < 40)$$

$$A = BH = 10 \times \frac{1}{40} = \frac{1}{4} \approx \underline{\underline{25\%}}$$

$\Rightarrow$ Calculate $P(10 < X < 26)$

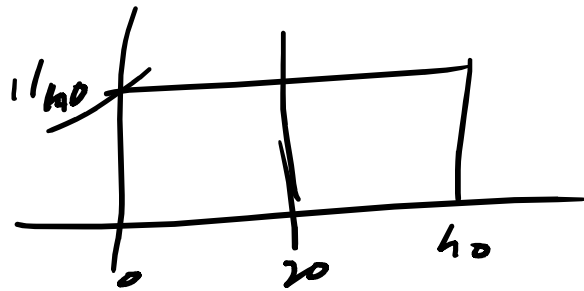


$$A = B \cdot h$$

$$= (26 - 10) \times \frac{1}{40} = \frac{16}{40} = \frac{4}{10} = \frac{2}{5}$$

$$= \underline{\underline{40\%}}$$

$$P(x = 20) = 0$$

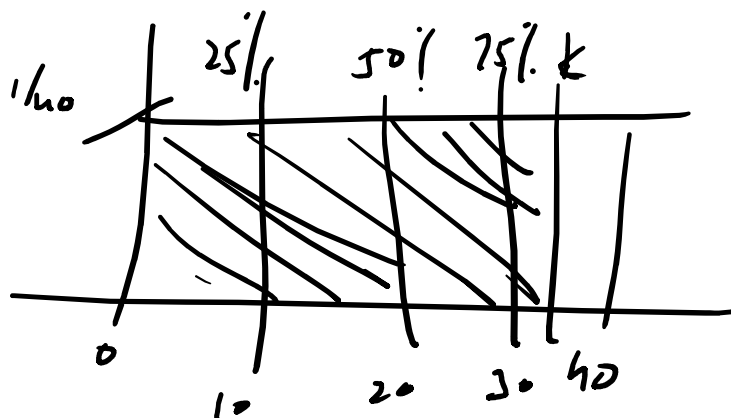


$$P(x > 45) = 0$$

$$\left\{ \begin{array}{l} \Rightarrow \text{Mean} = \boxed{\mu = \frac{a+b}{2}} = \frac{0+40}{2} = \underline{\underline{20}} \\ \Rightarrow \text{S.D} = \boxed{\sigma = \frac{b-a}{\sqrt{12}}} = \frac{40-0}{\sqrt{12}} = \underline{\underline{11.547}} \end{array} \right.$$

For uniform distⁿ only

=> What is 85th percentile?



$$A = BH$$

$$0.85 = (k - 0) \left(\frac{1}{h_0} \right)$$

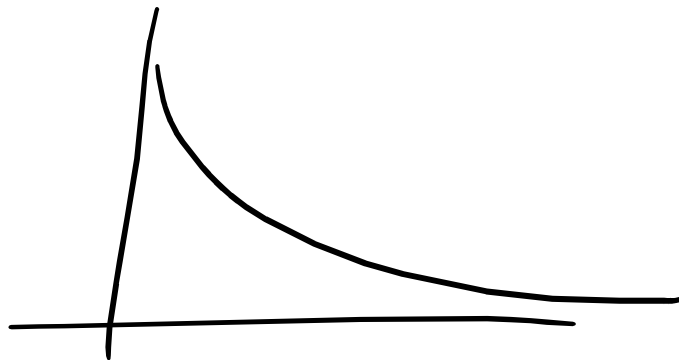
$$h_0 \times 0.85 = (k - 0) \times \left(\frac{1}{h_0} \right) \times h_0$$

$$k = \underline{\underline{34}}$$

=> Exponential Distribution:

$$f(x) = \lambda e^{-\lambda x}$$

$$\lambda = \frac{1}{\mu}$$



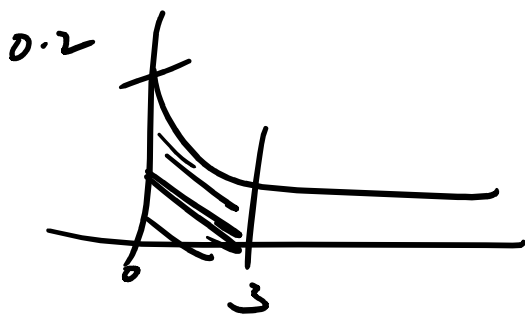
$\Rightarrow$ laptop with average life of 5 years.

$$\lambda = \frac{1}{u} = \frac{1}{5} = 0.20 \text{ years}^{-1}$$

$$u = 5 \text{ years} \quad e \approx 2.71 \dots$$

$$f(x) = \lambda e^{-\lambda x} = 0.20 \times e^{-0.20x}$$

$\Rightarrow$ What is prob that a laptop will last less than 3 years?



$$A_L = P(x < u) \\ = 1 - e^{-\lambda u}$$

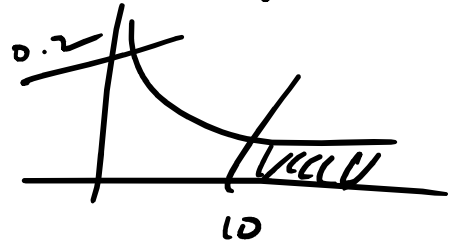
$$P(x < 3) = 1 - e^{-0.20(3)}$$

$$= 1 - e^{-0.6} = 0.4512 \approx \boxed{45.12\%}$$

$\Rightarrow$ Prob. that it will last more than 10 years?

→ prob. that it will live more than 10 years

$$A_R = P(X > 10) = e^{-\lambda x}$$



$$P(X > 10) = e^{-0.20(10)}$$

$$= e^{-2} = 0.1353 = \boxed{13.53\%}$$

⇒ Between 4 & 7 years.

$$P(4 < X < 7) = P(X < 7) - P(X < 4)$$



$$[1 - e^{-0.20(7)}] - [1 - e^{-0.20(4)}]$$

$$= 0.7534 - 0.55067$$

$$= \underline{0.20273} \approx \boxed{\underline{20.273\%}}$$